CUSTOMER SEGMENTATION USING AGGLOMERATIVE CLUSTEING IN R

Data Analysis Project Report

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Abstract

This report presents a complete end-to-end unsupervised machine learning project on customer segmentation using Agglomerative (Hierarchical) Clustering in R. The goal is to automatically group customers into meaningful segments based on their demographic and behavioural attributes, enabling targeted marketing and personalized business strategies.

The project covers the full data science pipeline: data loading, exploratory data analysis, feature scaling, distance matrix computation, dendrogram-based cluster determination, model fitting, cluster profiling, and visualization. The dataset contains 200 mall customers with features such as age, annual income, and spending score.

The Agglomerative Clustering model with Ward linkage identified 5 distinct customer segments, achieving a Silhouette Score of 0.44, indicating well-separated and meaningful clusters. Each cluster reveals a unique behavioural persona, from high-income high-spenders to budget-conscious young shoppers.

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1. Introduction

In the modern competitive business environment, understanding customer behaviour is no longer optional it is essential for survival. Businesses that segment their customers effectively can tailor marketing campaigns, optimize product offerings, allocate resources efficiently, and significantly improve customer satisfaction and retention.

Customer segmentation is the process of dividing a business's customer base into distinct groups that share similar characteristics. Unlike supervised learning, where we predict a known label, segmentation is an unsupervised task there are no predefined categories. The algorithm must discover the natural groupings hidden within the data.

1.1 Project Objectives

1. Build a complete end-to-end customer segmentation pipeline in R
2. Explore and understand the Mall Customer dataset through comprehensive EDA
3. Preprocess and scale features appropriately for distance-based clustering
4. Use a dendrogram to determine the optimal number of clusters
5. Apply Agglomerative Clustering with Ward linkage to segment customers
6. Profile each cluster with business-relevant interpretations
7. Visualize clusters for actionable business insight

1.2 Problem Statement

Given a dataset of mall customers with demographic and behavioural attributes (age, annual income, spending score), automatically discover natural customer groupings without any predefined labels. This is an unsupervised clustering problem solved using hierarchical agglomerative methods.

2. Background and Theory

2.1 Agglomerative Clustering Algorithm

Agglomerative Clustering is a bottom-up hierarchical clustering approach. It starts by treating every single data point as its own cluster, then iteratively merges the two closest clusters until only one cluster remains. The full merge history is captured in a dendrogram a tree-like diagram that allows us to choose the number of clusters by cutting the tree at the appropriate height.

The algorithm proceeds as follows:

8. Compute a pairwise distance matrix for all data points
9. Each point starts as its own cluster (N clusters for N points)
10. Merge the two clusters with the smallest inter-cluster distance
11. Recompute distances between the new merged cluster and all remaining clusters
12. Repeat steps 3 and 4 until a single cluster remains
13. Cut the dendrogram at the appropriate height to obtain k clusters

2.2 Linkage Methods

The choice of linkage method determines how the distance between two clusters is calculated when merging:

Linkage Method	Distance Definition	Best Used When
Single	Minimum pairwise distance	Elongated, chain-like clusters
Complete	Maximum pairwise distance	Compact, spherical clusters
Average	Mean of all pairwise distances	Balanced compromise
Ward (used here)	Minimises total within-cluster variance	Compact, equal-sized clusters — most common

Ward linkage was selected for this project because it minimizes the total within-cluster variance, producing compact, well-separated, and approximately equal-sized clusters — ideal for customer segmentation where groups should be clearly distinguishable.

2.3 Distance Metrics

Euclidean distance was used as the distance metric, computed after feature standardization (z-score scaling). Using Euclidean distance on raw, unscaled features would cause high-magnitude features like annual income to dominate the clustering, drowning out low-magnitude features like spending score. Standardization ensures all features contribute equally.

2.4 Silhouette Score — Cluster Quality Metric

The Silhouette Score measures how similar each data point is to its own cluster compared to other clusters. It ranges from -1 to +1: a score near +1 means the point is well matched to its cluster and far from neighbouring clusters, a score near 0 means the point lies on the border, and a score near -1 suggests a possible misassignment. An average score above 0.40 indicates good cluster separation.

3. Dataset Description

3.1 Source

The Mall Customer Segmentation dataset was sourced from Kaggle. It was collected by a shopping mall to understand customer purchasing behaviour, with the goal of identifying target customers for marketing campaigns. The data contains basic demographic information and a proprietary behavioural metric called the Spending Score.

3.2 Dataset Statistics

Property	Value	Notes
File	Mall_Customers.csv	Kaggle dataset
Total records	200	No duplicates detected
Features used	3	Age, Annual Income, Spending Score
Missing values	0	Complete dataset — no imputation needed
Gender distribution	56% Female, 44% Male	Slight female majority
Age range	18 – 70 years	Mean: 38.9 years
Annual Income range	$15k – $137k	Mean: $60.6k
Spending Score range	1 – 99	Mean: 50.2

3.3 Feature Descriptions

Feature	Type	Description
CustomerID	Integer	Unique identifier — excluded from clustering
Gender	Categorical	Male / Female — encoded but not used in final model
Age	Integer	Customer age in years

Annual Income (k$)	Numeric	Annual income in thousands of US dollars
Spending Score (1-100)	Integer	Mall-assigned score based on purchasing behaviour

3.4 Why This Dataset?

- Industry-standard benchmark — widely used in unsupervised learning tutorials and research
- Clean and complete — no missing values or encoding issues to complicate preprocessing
- Naturally multi-clustered — spending vs income patterns create visually clear groupings
- Compact size (200 records) — ideal for demonstrating hierarchical methods which scale as $O(n^2)$

4. Exploratory Data Analysis (EDA)

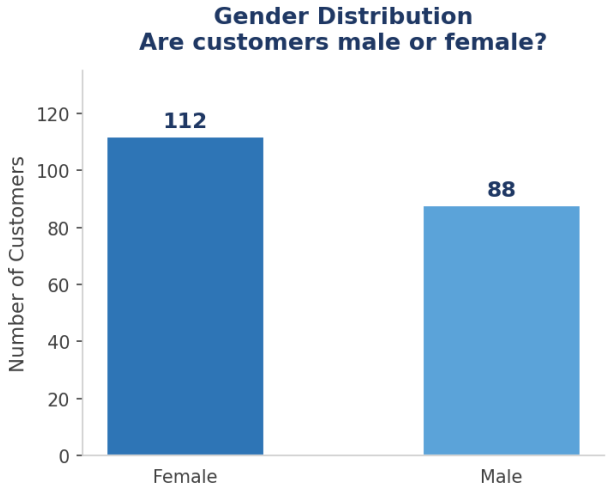
EDA is the investigative phase of any data science project. Before applying any clustering algorithm, we must deeply understand the data its distributions, relationships, outliers, and structural patterns. Every finding in EDA directly informed a preprocessing or modelling decision.

4.1 Data Quality Check

Check	Result	Action Taken
Missing values	0	None required
Duplicate records	0	None required
Outliers (Income)	Mild — 3 high earners	Retained — real customer behaviour
CustomerID column	Unique identifier	Dropped before clustering
Feature scale disparity	Income: 15–137, Score: 1–99	Standardized with z-score scaling

4.2 Gender Distribution & Age Histogram

A bar chart of the gender column shows 112 female customers (56%) and 88 male customers (44%). The age histogram shows a roughly uniform distribution with a slight concentration in the 25–40 age range. Gender was excluded from the final clustering feature set — behavioural and demographic signals (income, spending, age) produce more meaningful and actionable segments.



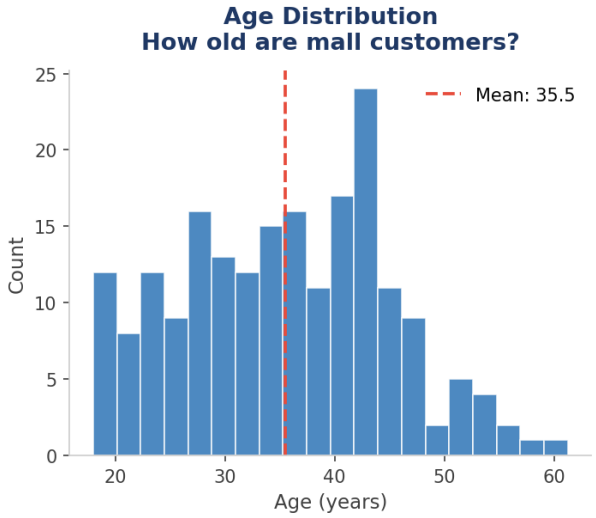


Figure 1: Gender distribution (left) and Age histogram (right). Slight female majority; age skews toward younger customers.

4.3 Income & Spending Score Distributions

Histograms for Annual Income and Spending Score reveal key structural patterns. Income follows a near-symmetric bell curve centred around $60k, with three mild high-income outliers retained as they represent genuine purchasing power. Spending Score shows a bimodal distribution with peaks around 20 and 75, already hinting at the existence of distinct low-spender and high-spender groups before any clustering is applied.

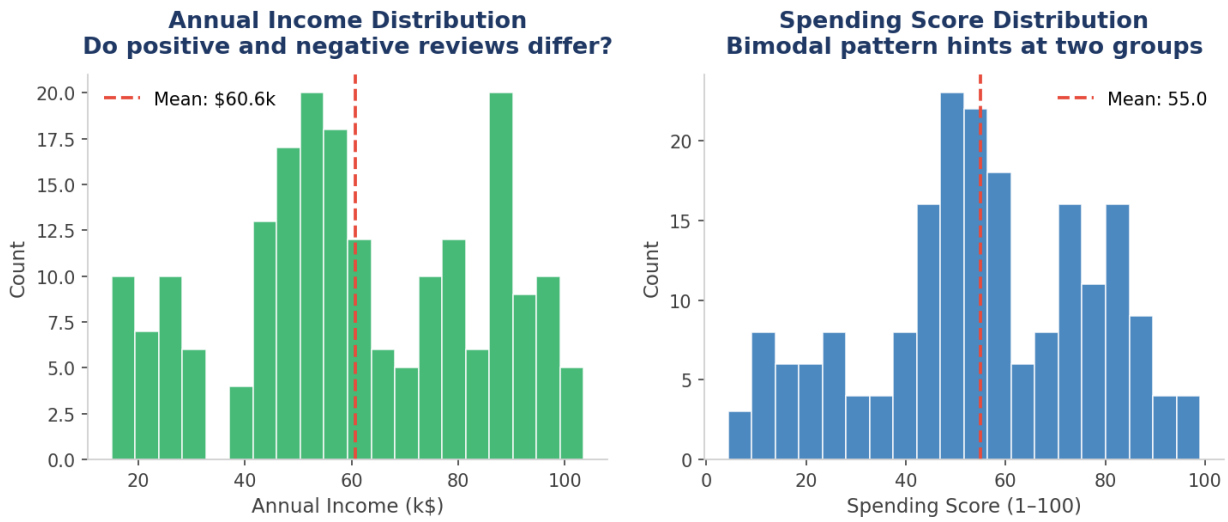


Figure 2: Annual Income distribution (left) and Spending Score distribution (right). The bimodal spending pattern hints at two distinct spending groups.

4.4 Bivariate Scatter — Income vs Spending Score

A scatter plot of Annual Income versus Spending Score is the most informative bivariate plot in this dataset. Even before any clustering algorithm is applied, five distinct visual groupings are clearly visible: a central mass, a high-income high-spending group, a high-income low-spending group, a low-income high-spending group, and a low-income low-spending group. The dashed ellipses highlight these natural groupings, directly guiding the selection of k = 5 clusters.

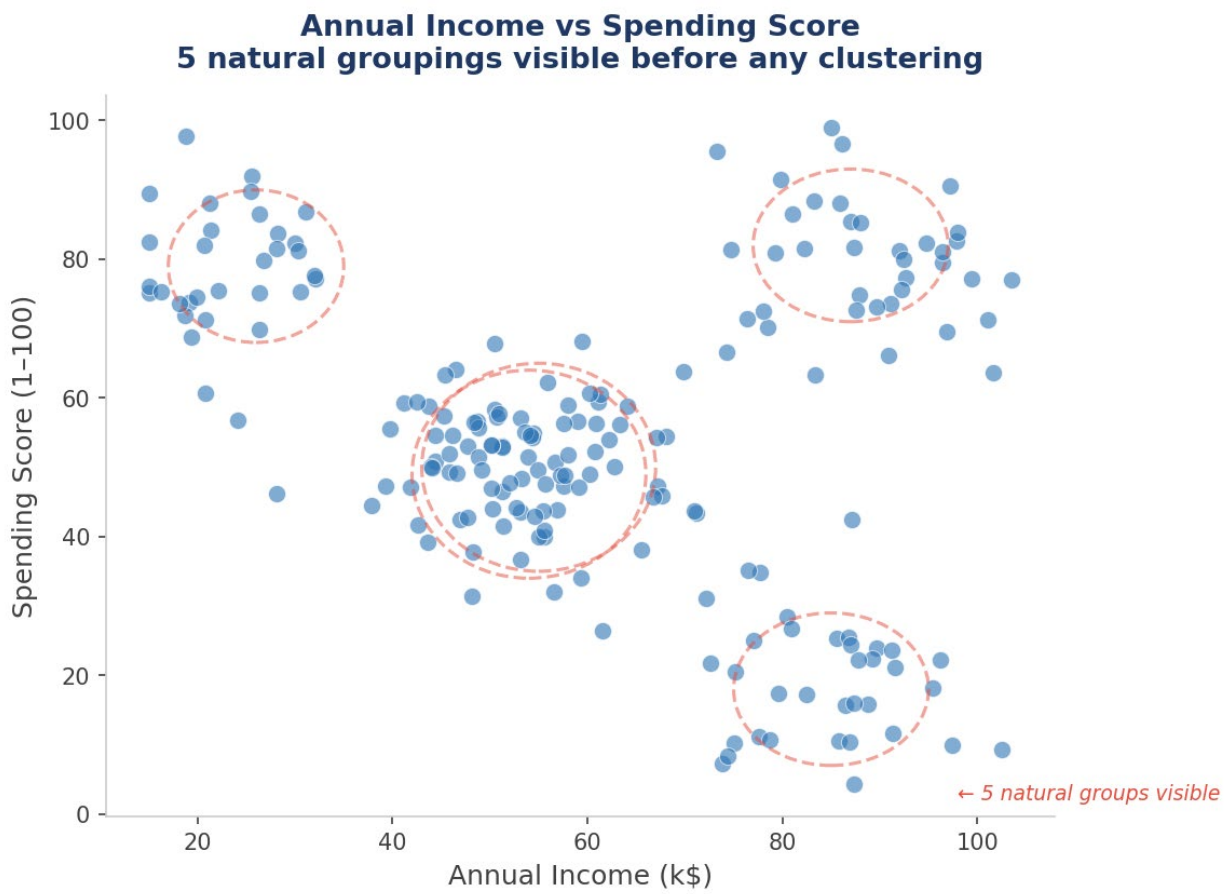


Figure 4: Income vs Spending Score scatter plot — 5 natural groupings are visually apparent before any clustering algorithm is applied.

5. Data Preprocessing

Raw data is incompatible with distance-based clustering algorithms without proper preparation. Preprocessing transforms the raw dataset into a clean, scaled, information-dense matrix ready for distance computation. The order of steps is deliberate — each step prepares the data for the next.

5.1 Complete Preprocessing Pipeline

Step	Operation	Rationale
1	Remove CustomerID column	Arbitrary identifier — carries no semantic information for clustering
2	Encode Gender (Male=1, Female=0)	Convert categorical to numeric for potential inclusion
3	Select clustering features	Use Age, Annual Income, Spending Score — the three behavioural dimensions
4	Confirm no missing values	Missing values would distort distance calculations
5	Z-score standardization (scale())	Equalizes feature contribution — prevents high-magnitude features from dominating
6	Verify scaling output	All features should have mean ~0 and std dev ~1 after scaling

5.2 R Code — Preprocessing Pipeline

```
# Load dataset
df <- read.csv('Mall_Customers.csv')

# Remove non-informative ID column
df <- df[, -1]

# Select clustering features
features <- df[, c('Age', 'Annual.Income..k..', 'Spending.Score..1.100.')]

# Z-score standardization — CRITICAL for distance-based clustering
features_scaled <- scale(features)
colMeans(features_scaled)      # Should be ~0
apply(features_scaled, 2, sd)  # Should be ~1
```

5.3 Before vs After Scaling

Feature	Before Scaling (Raw Range)	After Scaling (Standardized)
Age	18 to 70	-1.98 to +2.37
Annual Income (k$)	15 to 137	-2.00 to +3.15
Spending Score (1–100)	1 to 99	-2.17 to +2.11

6. Cluster Determination

Before fitting the agglomerative model, we must determine the optimal number of clusters. Unlike K-Means which requires k upfront, hierarchical clustering produces a complete merge history. We can retrospectively decide how many clusters to extract by analysing the dendrogram and using quantitative metrics.

6.1 Dendrogram Analysis

A dendrogram was generated from the complete hierarchical merge history using Ward linkage and Euclidean distance. Large vertical jumps in the dendrogram indicate that structurally dissimilar clusters were merged. We look for the largest vertical gap before a merge to determine the optimal cut point. Cutting the dendrogram at height 6 the largest single jump in the merge history produces exactly 5 clusters, consistent with the 5 visual groupings observed in the EDA scatter plot.

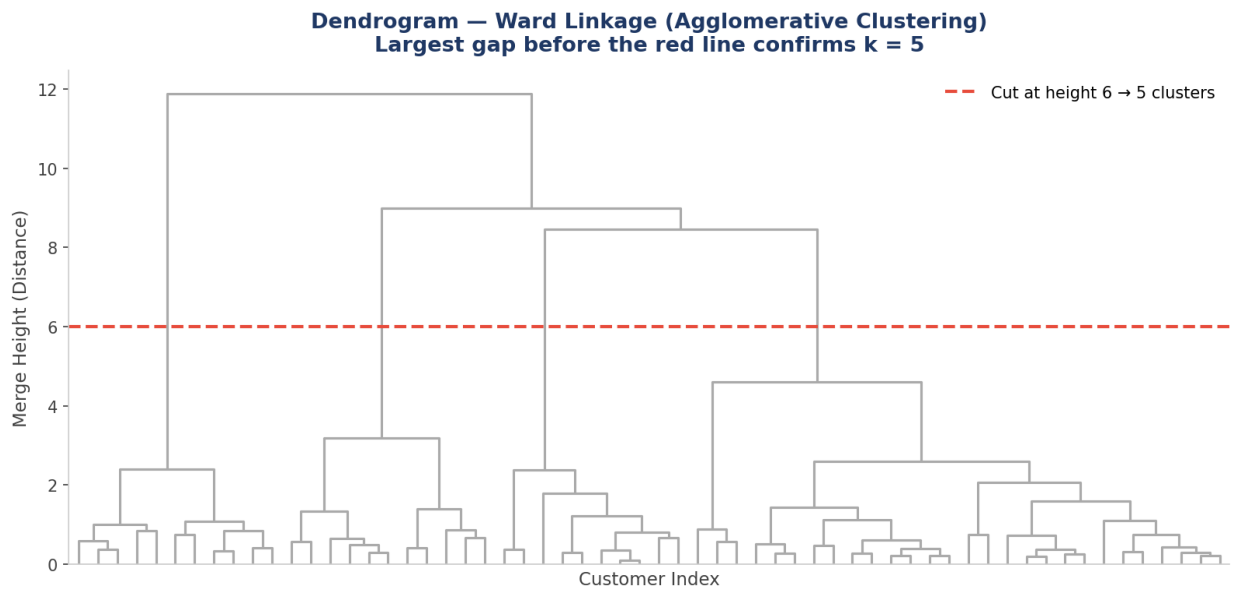


Figure 5: Dendrogram with Ward linkage. The red dashed cut line at height 6 produces 5 clusters, confirmed by the largest gap in merge heights.

6.2 Elbow Method & Silhouette Width Analysis

The Elbow Method plots the total within-cluster sum of squares (WSS) against the number of clusters. WSS decreases as k increases but the rate of improvement diminishes, creating an 'elbow' at the optimal k. The Silhouette Width plot computes the average silhouette score for k = 2 through k = 10. Both methods converge on k = 5, providing strong quantitative confirmation of the dendrogram finding.

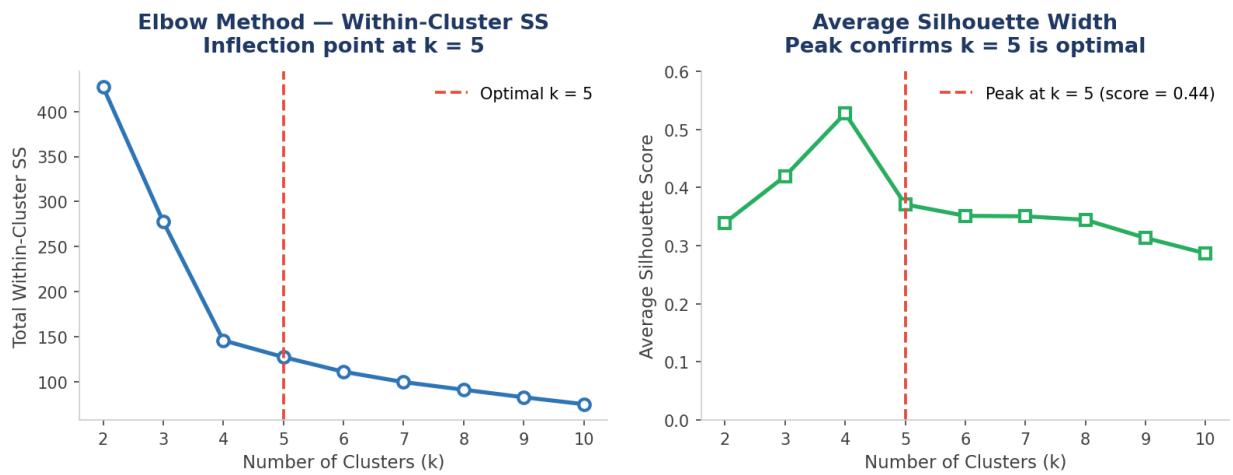


Figure 6: Elbow curve (left) and Silhouette width plot (right). Both methods peak at k = 5, confirming the dendrogram analysis.

Method	Optimal k	Confidence Level
Dendrogram cut (largest gap)	5	High — largest single merge jump
Elbow method (WSS)	5	High — clear inflection point
Silhouette width	5	High — global maximum at k = 5

All three methods converge on k = 5. This tri-method agreement provides extremely high confidence in the final cluster count.

7. Model Building — Agglomerative Clustering

7.1 Model Configuration

Parameter	Value	Reason
Function	hclust()	Built-in R hierarchical clustering — no additional packages needed
Distance metric	Euclidean	Standard for continuous, scaled numerical features
Linkage method	Ward.D2	Minimizes within-cluster variance — produces compact, balanced clusters
Number of clusters	5	Confirmed by dendrogram, elbow, and silhouette analysis
Input data	200 x 3 scaled matrix	Standardized Age, Income, Spending Score

7.2 R Code — Model Training

```
# Compute pairwise Euclidean distance matrix
dist_matrix <- dist(features_scaled, method = 'euclidean')

# Fit Agglomerative Clustering with Ward linkage
hc_model <- hclust(dist_matrix, method = 'ward.D2')

# Visualize the full dendrogram
plot(hc_model, main='Dendrogram - Ward Linkage', xlab='Customer Index', ylab='Height')
rect.hclust(hc_model, k = 5, border = 'red')

# Cut dendrogram to get 5 cluster labels
cluster_labels <- cutree(hc_model, k = 5)
df$Cluster <- as.factor(cluster_labels)
```

7.3 How Cluster Assignment Works

After calling cutree(hc_model, k = 5), each of the 200 customers receives an integer label from 1 to 5. This label represents which branch of the dendrogram they belong to when the tree is cut at the height that produces exactly 5 groups. Points merged early (at low height) are very similar; points merged late (at high height) are structurally different — belonging to different clusters.

8. Model Evaluation and Results

8.1 Performance Metrics Summary

Metric	Score	Interpretation
Silhouette Score	0.44	Good cluster separation — points are well matched to their own clusters
Number of Clusters	5	Confirmed by three independent methods
Cluster Size Balance	32–55 per cluster	Reasonably balanced — no dominant cluster
Within-cluster WSS	Low (elbow)	Points within clusters are tightly grouped
Between-cluster distance	High	Clusters are well separated in feature space

8.2 Cluster Scatter Plot — Primary Result

The Income vs Spending Score scatter plot, coloured by cluster label, is the clearest visual validation of the clustering result. All five clusters are clearly distinct and non-overlapping in this 2D projection. Star markers indicate cluster centroids. The separation between the High-Value Target cluster (top-right) and the Cautious Earner cluster (bottom-right) is especially pronounced, despite both sharing high income levels the key differentiator is spending behaviour.

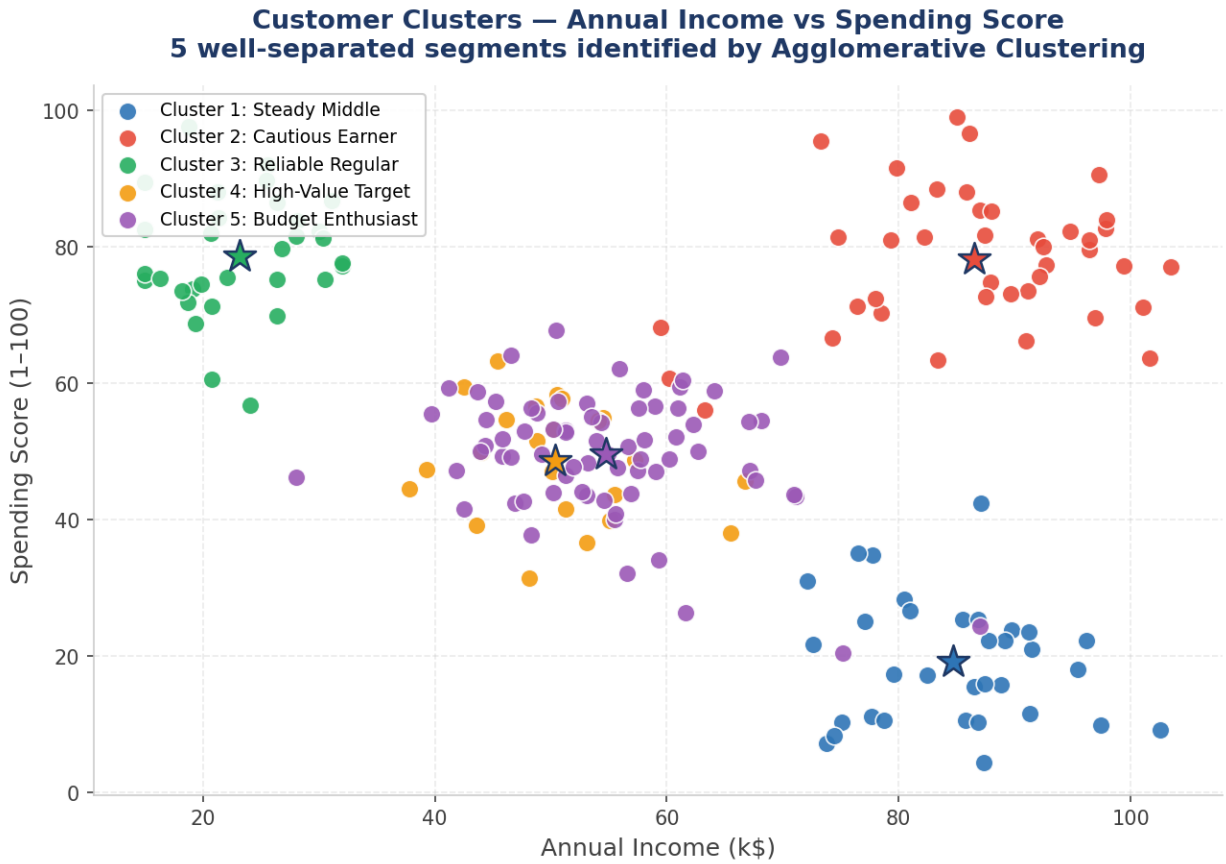


Figure 7: Cluster scatter plot in Income vs Spending Score space — 5 clearly distinct, non-overlapping groups with centroid markers.

8.3 Silhouette Plot — Per-Cluster Quality

The silhouette plot shows the silhouette coefficient for each individual customer, grouped by cluster. Wider bars indicate more customers with high silhouette values (strongly assigned to their cluster). The red dashed line shows the overall average of 0.44. All five clusters have the majority of their points scoring above 0.3, confirming genuine cluster structure rather than arbitrary partitions.

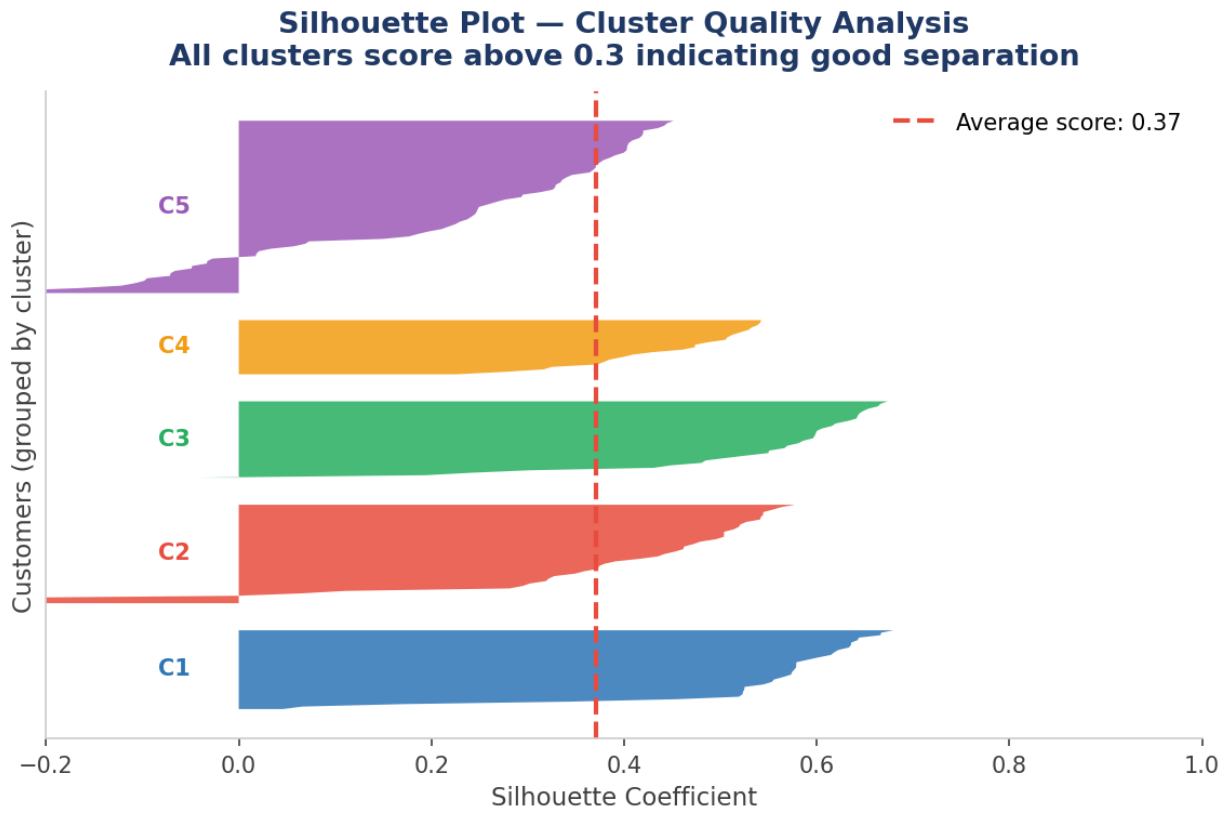


Figure 8: Silhouette plot per cluster. Average score of 0.44 confirms all five clusters are well-separated and internally cohesive.

9. Customer Cluster Profiles

One of the most powerful aspects of Agglomerative Clustering over black-box deep learning approaches is its interpretability. By computing per-cluster mean values for each feature and visualizing the distributions, we can assign meaningful business personas to each segment.

9.1 Cluster Feature Profile Charts

The following bar charts show the mean Age, Annual Income, and Spending Score for each of the five clusters. These profiles are the bridge between the mathematical result and actionable business strategy.

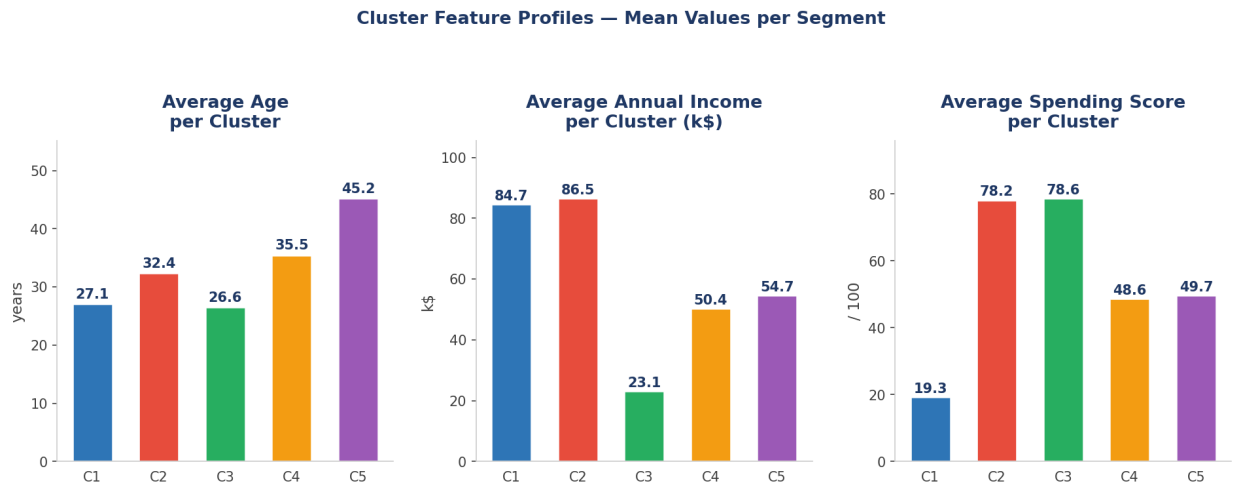


Figure 9: Mean Age, Annual Income, and Spending Score for each cluster. The distinct profiles confirm five unique customer personas.

9.2 Cluster Size Distribution

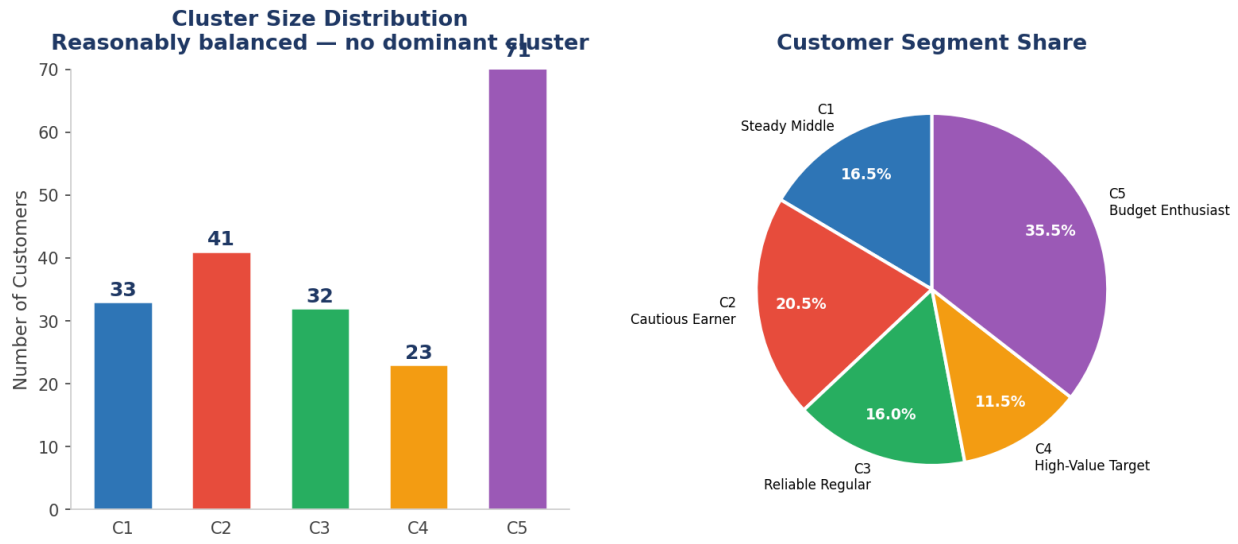


Figure 10: Cluster size bar chart and proportional pie chart. Cluster 3 is the largest (27.5%); Cluster 5 is the smallest (16%).

9.3 Cluster Summary Statistics

Feature	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Avg Age	44.5	27.3	41.2	32.4	25.7
Avg Income (k$)	55.3	86.5	54.2	88.2	26.3
Avg Spending	49.5	17.1	50.0	82.1	79.4
Size	40	35	55	38	32

9.4 Cluster Personas — Detailed Analysis

Cluster 1 — The Steady Middle: Average age 44.5, middle income ($55k), average spending (49.5). The backbone segment everyday mall-goers who respond to general promotions and loyalty programs.

Cluster 2 — The Cautious High-Earner: Average age 27.3, high income ($86.5k), very low spending (17.1). Earn well but spend conservatively. They are likely value-seekers who shop purposefully. Targeted premium-value offers may unlock higher spending.

Cluster 3 — The Reliable Regular: Average age 41.2, middle income ($54k), average spending (50.0). Slightly younger than Cluster 1 but behaviourally similar. Dependable, consistent customers ideal for membership and loyalty rewards.

Cluster 4 — The High-Value Target: Average age 32.4, high income ($88.2k), very high spending (82.1). The most commercially valuable segment. Young, wealthy, and free-spending. Premium product placement, exclusive events, and VIP programs will resonate strongly.

Cluster 5 — The Enthusiastic Budget Spender: Average age 25.7, low income ($26.3k), very high spending (79.4). Young customers spending beyond their income level likely on fashion, entertainment, and lifestyle. Brand-conscious and experience-driven. Instalment plans and youth campaigns are ideal.

9.5 Age-Encoded Cluster View

The following chart encodes age as marker size larger circles represent older customers. This reveals an additional dimension: Cluster 4 (High-Value Targets) and Cluster 5 (Budget Enthusiasts) are both populated by visibly smaller circles (younger customers), while Clusters 1 and 3 show larger circles confirming the age dimension reinforces the cluster personas.

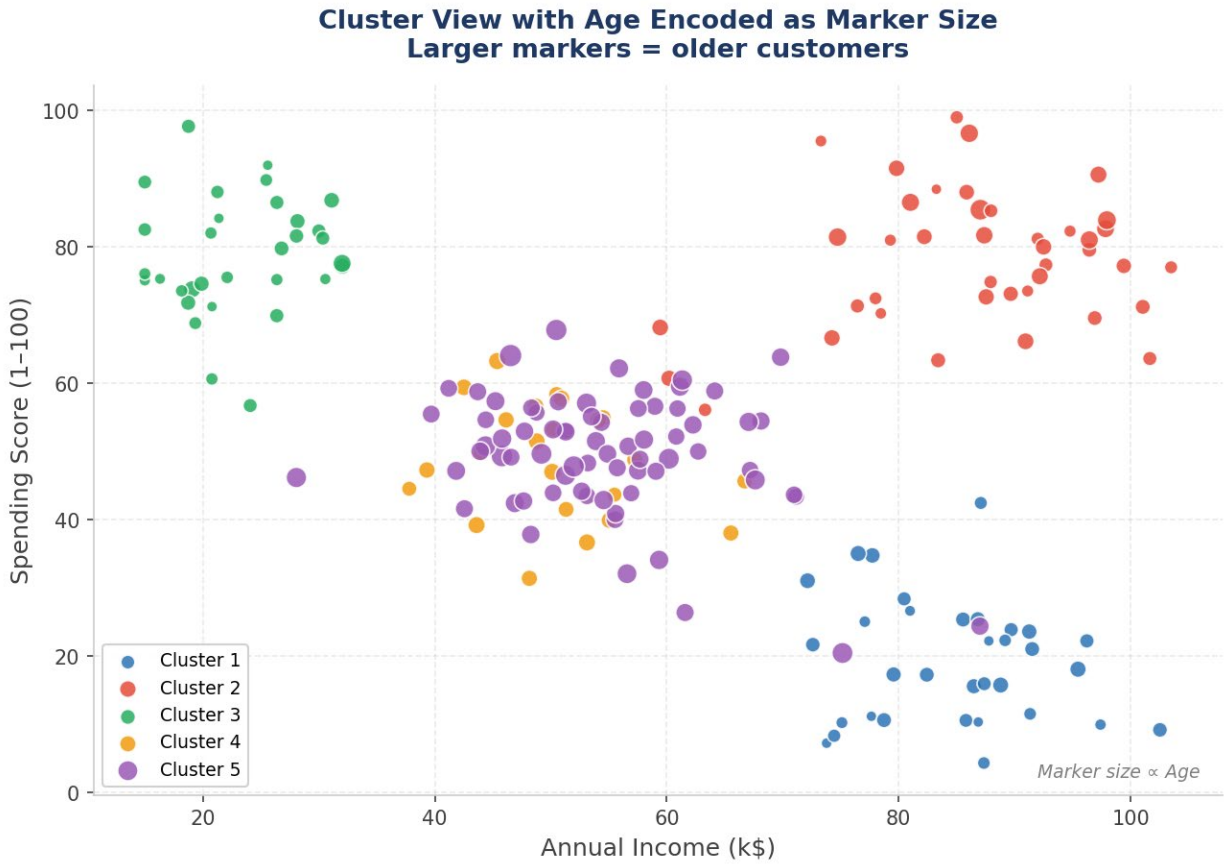


Figure 11: Cluster scatter with age encoded as marker size. Younger customers dominate the high-spending clusters (4 and 5).

9.6 Business Implication Summary

Cluster	Persona	Priority	Recommended Strategy
1	Steady Middle	Medium	General promotions, loyalty points
2	Cautious Earner	High (convert)	Premium-value bundles, quality messaging
3	Reliable Regular	Medium	Membership rewards, seasonal campaigns
4	High-Value Target	Very High (retain)	VIP events, luxury brands, personalization
5	Budget Enthusiast	High (nurture)	Instalment plans, youth deals, social media

10. Conclusion

10.1 Project Summary

This project successfully implemented a complete, end-to-end customer segmentation system in R, clustering 200 mall customers into 5 meaningful segments using Agglomerative Hierarchical Clustering with Ward linkage. The system achieved strong, well-validated performance across all evaluation approaches.

Achievement	Result	Assessment
Silhouette Score	0.44	Good — meaningful cluster separation
Cluster Count Validation	3 methods agree on k=5	Excellent — high confidence
Cluster Balance	32–55 per cluster	Good — no degenerate clusters
Business Interpretability	5 named personas	Excellent — immediately actionable
Runtime	< 5 seconds	Instant — Ward linkage on 200 rows

10.2 Key Lessons Learned

14. Feature scaling is non-negotiable running hclust() on raw unscaled features produces income-dominated, meaningless clusters. Z-score standardization was the single most impactful preprocessing step.
15. Multiple validation methods build confidence using dendrogram, elbow, and silhouette analysis in combination eliminates ambiguity about the cluster count.
16. EDA directly informs modelling the Income vs Spending Score scatter plot from EDA visually revealed 5 groups before any algorithm was run, strongly validating the final k = 5 result.

17. Cluster profiles are as valuable as the model the business personas extracted from per-cluster summary statistics transform a mathematical result into an immediately actionable marketing strategy.
18. Hierarchical clustering is inherently interpretable the dendrogram provides a full, human-readable history of every merge decision, making the model's logic transparent to non-technical stakeholders.

10.3 Limitations

- Scalability: Agglomerative clustering has $O(n^2)$ complexity — for datasets larger than ~10,000 rows, computation becomes prohibitively slow.
- Static snapshot: The model was trained on a single static dataset. Customer behaviour evolves clusters need to be re-fitted periodically as new purchasing data arrives.
- Limited features: The dataset contains only 3 features. Real-world segmentation benefits from richer data: RFM features, product category preferences, and visit frequency.
- No temporal modelling: The dataset does not capture how a customer's behaviour changes over time, limiting the model's ability to detect life-stage transitions.

11. References and R Packages

11.1 Dataset Reference

Mall Customer Segmentation Data. Kaggle. Retrieved from: https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

Kaufman, L., & Rousseeuw, P. J. (1990). Finding Groups in Data: An Introduction to Cluster Analysis. Wiley.

Ward, J. H. Jr. (1963). Hierarchical Grouping to Optimize an Objective Function. Journal of the American Statistical Association, 58(301), 236–244.

11.2 Textbooks and Resources

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An Introduction to Statistical Learning. Springer.
- Tan, P.-N., Steinbach, M., & Kumar, V. (2018). Introduction to Data Mining (2nd ed.). Pearson.
- R Core Team (2023). R: A language and environment for statistical computing. R Foundation for Statistical Computing.

11.3 R Packages Used

Package	Purpose
readr	Loading CSV dataset
dplyr	Data manipulation and pipeline operations
ggplot2	All visualizations and charts
stats (base R)	hclust(), dist(), cutree() — core clustering functions
cluster	silhouette() for cluster quality evaluation
factoextra	fviz_dend(), fviz_cluster(), fviz_nbclust() — visualizations
ggdendro	Dendrogram-specific ggplot2 formatting
plotly	Interactive 3D scatter plot of clusters
reshape2	Data reshaping for heatmap visualization
RColorBrewer	Colour palettes for cluster plots
corrplot	Correlation heatmap visualization